README

Bengali NLP Safety Challenge 2026
Targeted Upgrade of the 0.591 Baseline

Project overview and run guide for **NLP-02_Tokenizing_Our_Tears_Zikra.ipynb**

1. Project overview

This notebook develops a multi-label Bengali toxicity classifier for the BUBT Bengali NLP Safety Challenge 2026. For each comment, it predicts seven binary labels: overall toxicity and six toxicity attributes. The notebook is designed as a controlled upgrade of an earlier 0.591-scoring baseline; its stated goal of 0.61+ is a target, not a guaranteed result.

2. Labels and evaluation

Label	Meaning
is_toxic	Whether the comment is toxic overall
implicit_toxicity	Indirect or implied toxicity
sarcasm	Sarcastic toxic expression
contains_profanity	Profanity or abusive words
threatening	Threatening language
identity_attack	Attack on a person or group identity
sexual_content	Sexual content

The competition metric averages two components: macro-F1 for is_toxic and the mean positive-class F1 across the six attributes. The notebook tunes thresholds against this combined score rather than relying on accuracy alone.

3. Method at a glance

• **Transformer branch:** pretrained BanglaBERT-base and BanglaBERT-large with a bidirectional LSTM head and masked mean pooling.

• **Lexical branch:** word- and character-level TF-IDF features with logistic regression to capture spelling variants, Banglish, slang, and profanity patterns.

• **Training:** five-fold multilabel-stratified cross-validation, focal-style loss, and clipped positive-class weighting for imbalanced labels.

• **Duplicate handling:** diagnostic checks for repeated texts; optional soft targets for duplicate groups with conflicting labels.

• **Decision rules:** out-of-fold (OOF) search for ensemble weights, per-label thresholds, and an overall-toxicity gate.

• **Final inference:** retrain selected transformer variants on the combined training and validation data, then produce predictions for the competition test set.

4. Requirements and data

Run the notebook in an environment with Python, a compatible PyTorch installation, Transformers, scikit-learn, pandas, NumPy, and iterative-stratification. The notebook attempts to install iterative-stratification and scikit-learn. A CUDA GPU is recommended because transformer fine-tuning can be slow on CPU.

Provide the official competition data files in the format expected by the notebook. The notebook includes data-loading logic intended for Kaggle and Google Colab and expects labeled training data, a labeled validation split, and an unlabeled test split. Check the data-loading cell and column names before running.

5. How to run

1. Open the notebook in Kaggle or Google Colab.
2. Attach or upload the official competition dataset; confirm that the train, validation, and test files are detected correctly.
3. Run the notebook from top to bottom. Allow the tokenization audit, diagnostics, model training, cross-validation, and ensemble-search cells to finish.
4. Review the printed OOF scores, per-label diagnostics, selected ensemble weights, thresholds, and toxicity gate.
5. Run the final training/inference section to generate the submission CSV.
6. Verify that the CSV contains the required ID column and seven binary prediction columns, with the expected row count, before submitting it to the competition.

6. Outputs to record

After each run, record the OOF score for BanglaBERT-base, the OOF score for BanglaBERT-large, the best ensemble OOF score, and the Kaggle public leaderboard score. Save the per-label error analysis and the exact configuration used so results can be reproduced.

7. Important limitations and checks

• The notebook’s stated 0.61+ target is aspirational; actual performance depends on the data, environment, random variation, and evaluation.

• OOF-based ensemble weights and thresholds can overfit the same OOF predictions used to select them. Treat the selected OOF score as a model-selection estimate, not a fully unbiased final estimate.

• Repeated comments can leak across row-level folds unless duplicate groups are kept together. Consider grouped cross-validation by normalized text for a more reliable evaluation.

• Check that duplicate-aware target handling is consistent between cross-validation and final training.

• Do not report unverified leaderboard scores or claim improvements until they are confirmed by an actual run.

8. Suggested citation / attribution

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